EMBEDDED SYSTEM DESIGN

Smart Irrigation System

Project Topic:

Automated Smart Irrigation System
for a Vegetable Farm in Rwanda

Module:	EMBEDDED SYSTEM DEVELOPMENT
System:	Smart Irrigation System
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1. Introduction

Agriculture is an important sector in Rwanda, and effective water management is essential for maintaining good crop production. The farmer in this scenario owns a large vegetable farm and experiences difficulties in determining when crops require water. At some times, the crops receive excessive amounts of water, while at other times they remain dry because irrigation is not performed on time.

These problems result in water wastage, increased labour costs and reduced crop production. To address these challenges, the farmer has proposed the introduction of a Smart Irrigation System.

The proposed system is an embedded system that automatically monitors the moisture level of the soil and controls water supply according to the soil condition. A soil moisture sensor collects information from the soil and sends the information to a microcontroller. The microcontroller processes the sensor readings and determines whether irrigation is required. When the soil is too dry, the microcontroller activates a water pump through a relay module. When sufficient moisture is detected, the pump is switched off.

The system also includes an LCD display that allows the farmer to view the soil moisture condition and the status of the water pump. The design aims to reduce water wastage, reduce unnecessary labour, improve irrigation timing and use electrical power efficiently.

2. System Objectives

The main objectives of the proposed Smart Irrigation System are:

1. To automatically monitor the moisture level of agricultural soil.
2. To determine when irrigation is required based on soil moisture.
3. To automatically control a water pump.
4. To reduce unnecessary water consumption.
5. To reduce the labour required for manual irrigation.
6. To provide the farmer with clear information about soil moisture.
7. To improve the reliability and efficiency of the irrigation process.
8. To use electrical power efficiently by operating the pump only when necessary.

3. Task 1: Analysis of System Requirements

System requirements describe what the proposed Smart Irrigation System must do and how it should operate.

Requirements can be divided into two major categories:

- Functional requirements
- Non-functional requirements

3.1 Functional Requirements

Functional requirements describe the specific functions or activities that the system must perform.

The identified functional requirements are shown in Table 1.

Table 1: Functional Requirements

ID	Functional Requirement
FR1	The system shall measure the moisture level of the soil.
FR2	The system shall send soil moisture readings to the microcontroller.
FR3	The microcontroller shall compare the moisture reading with a predefined moisture threshold.
FR4	The system shall automatically turn on the water pump when the soil is too dry.
FR5	The system shall turn off the water pump when sufficient soil moisture is detected.
FR6	The system shall display soil moisture information to the farmer.
FR7	The system shall continuously or periodically monitor the soil moisture level while operating.
FR8	The system shall automatically control irrigation without requiring continuous manual operation.

3.2 Non-Functional Requirements

Non-functional requirements describe how the system should operate and the qualities that the system should possess.

Table 2: Non-Functional Requirements

ID	Non-Functional Requirement
NFR1	The system should operate reliably for long periods.
NFR2	The system should use electrical power efficiently.
NFR3	The system should respond quickly when the soil becomes too dry.
NFR4	The system should be simple for the farmer to understand and operate.
NFR5	The system should be sufficiently durable for use in a farm environment.
NFR6	The system should provide accurate and consistent soil moisture measurements.

3.3 Difference Between Functional and Non-Functional Requirements

Functional and non-functional requirements describe different aspects of a system.

Table 3: Difference Between Functional and Non-Functional Requirements

Functional Requirements	Non-Functional Requirements
Describe what the system must do.	Describe how well the system should operate.
Focus on system functions and behaviour.	Focus on performance, reliability, efficiency, usability and other qualities.
Example: The system shall turn on the pump when the soil is dry.	Example: The system should respond quickly when dry soil is detected.
Example: The system shall display soil moisture information.	Example: The displayed information should be clear and easy for the farmer to understand.

4. Task 2: Identification and Selection of System Components

Based on the identified requirements, suitable hardware and software components are selected for the Smart Irrigation System.

The main components are:

1. Arduino Uno microcontroller
2. Capacitive soil moisture sensor
3. Relay module
4. DC water pump
5. Appropriate power supply
6. LCD display
7. Connecting wires and irrigation pipes
8. Embedded software/firmware

4.1 Arduino Uno Microcontroller

The Arduino Uno is selected as the main microcontroller for the prototype.

Its main functions include:

- Receiving readings from the soil moisture sensor.
- Processing the sensor readings.
- Comparing the readings with predefined moisture thresholds.
- Controlling the relay module.
- Controlling the water pump indirectly through the relay.
- Sending information to the LCD display.

The Arduino Uno is suitable because it is inexpensive, easy to program, widely available and provides both analog input and digital output pins required by the system.

4.2 Capacitive Soil Moisture Sensor

The soil moisture sensor is responsible for measuring the moisture condition of the soil.

The sensor provides an electrical signal that changes according to the moisture level. The Arduino reads this signal and uses it to determine whether the soil requires irrigation.

A capacitive soil moisture sensor is suitable for a prototype because it is designed for soil moisture measurement and is less prone to electrode corrosion than basic resistive sensors.

4.3 Relay Module

The relay module acts as an electrically controlled switch between the microcontroller and the water pump.

The Arduino cannot directly provide the current required by a typical water pump. Therefore, the Arduino sends a low-power control signal to the relay, while the relay switches the pump's separate power circuit.

The basic operation is:

$$\textbf{Arduino} \rightarrow \textbf{Relay ON} \rightarrow \textbf{Pump ON}$$

and:

$$\textbf{Arduino} \rightarrow \textbf{Relay OFF} \rightarrow \textbf{Pump OFF}$$

The relay must be appropriately rated for the pump's voltage and current.

4.4 Water Pump

The water pump supplies water to the crops.

When the Arduino determines that the soil is too dry, it activates the relay and the pump starts operating. The pump transfers water from the water source to the irrigation system.

The pump should have an appropriate power supply and should not be powered directly from an Arduino output pin.

4.5 Power Supply

The power supply provides electrical energy to the system.

The Arduino, sensor, LCD and relay require appropriate regulated power. The pump should normally have a separate appropriately rated supply because it can require significantly more current than the microcontroller can provide.

Possible power sources for a future farm installation include:

- AC mains with a suitable regulated supply
- Battery
- Solar panel with a charge controller and battery

For the prototype, an appropriate regulated DC power supply can be used.

4.6 LCD Display

A 16×2 LCD display can be used to provide information to the farmer.

For example, the display can show:

```
Soil Moisture: 25%
Status: DRY
Pump: ON
```

When the soil becomes sufficiently moist, it could display:

```
Soil Moisture: 48%
Status: OK
Pump: OFF
```

The LCD makes the system easier to monitor without requiring a computer.

4.7 Communication Module

A communication module is not required for the basic version of the system because the farmer only needs to view the information locally through the LCD display.

However, the system could be upgraded in the future using an ESP32 microcontroller, which includes Wi-Fi and Bluetooth connectivity.

A future version could provide:

- Smartphone monitoring
- Remote notifications
- Cloud data storage
- Remote irrigation control

For the initial prototype, Arduino Uno is preferred because it keeps the design simple and suitable for an embedded-system prototype.

4.8 Embedded Software/Firmware

The embedded software is programmed into the Arduino microcontroller.

The firmware is responsible for:

- Reading the soil moisture sensor.
- Converting or interpreting the sensor reading.
- Comparing the reading with irrigation thresholds.
- Controlling the relay.
- Updating the LCD.
- Repeating the monitoring process.

A simple control algorithm is shown below.

4.9 Control Algorithm

The embedded software continuously monitors the soil moisture level and automatically controls the irrigation pump according to the measured soil condition.

The control algorithm is illustrated in Figure 4.9.

Smart Irrigation System Control Algorithm

The algorithm operates continuously. The soil moisture sensor measures the moisture condition of the soil, and the microcontroller processes the sensor reading before displaying the moisture level. The microcontroller then determines whether the soil is too dry. If the soil requires irrigation, the pump is switched ON. Otherwise, the pump remains OFF. After a short interval, the system repeats the monitoring process to maintain the required soil moisture condition.

For a practical implementation, separate ON and OFF thresholds can be used. For example, the pump could turn ON below 30% moisture and turn OFF above 45%. This reduces rapid switching when the reading fluctuates around a single threshold.

4.10 Summary of Selected Components

Table 4: Selected Components and Their Functions

Component	Purpose
Arduino Uno	Main microcontroller that processes sensor data and controls the system.
Capacitive Soil Moisture Sensor	Measures the moisture condition of the soil.
Relay Module	Electrically switches the water pump on or off.
Water Pump	Supplies water to the crops.
Power Supply	Provides appropriate electrical power to the system components.
16×2 LCD	Displays soil moisture and pump status to the farmer.
Embedded Firmware	Provides the logic for monitoring and automatic irrigation control.
Irrigation Pipes	Transport water from the pump to the crops.

5. Task 3: System Block Diagram

The block diagram below shows the major components of the Smart Irrigation System and the direction of data and control signals.

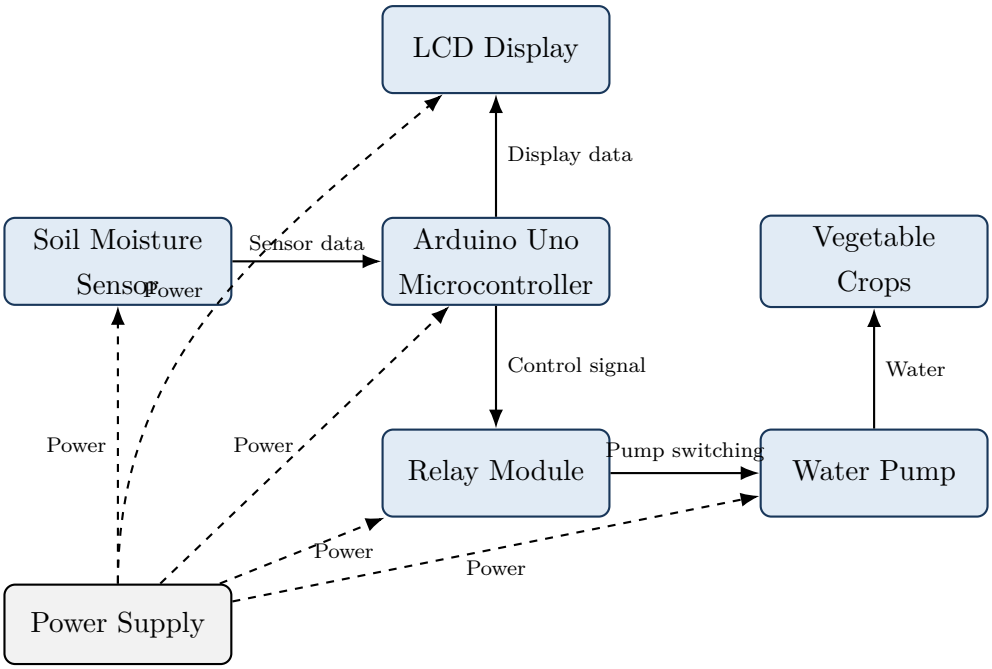


Figure 1: Smart Irrigation System Block Diagram

5.1 Explanation of the Block Diagram

The operation of the system shown in Figure 1 can be explained as follows:

1. The soil moisture sensor measures the moisture condition of the soil.
2. The sensor sends its reading to the Arduino Uno.
3. The Arduino processes the reading and compares it with the predefined moisture threshold.
4. The Arduino sends the current soil moisture information to the LCD display.
5. If the soil is too dry, the Arduino sends a control signal to the relay module.
6. The relay switches the water pump on.
7. The water pump supplies water to the crops.
8. The soil moisture increases as irrigation takes place.
9. The sensor measures the soil again.
10. When sufficient moisture is detected, the Arduino switches the relay off and the pump stops.

6. Task 4: Requirement-to-Component Mapping

The requirement-to-component mapping demonstrates how each identified requirement will be implemented by the selected hardware or software component.

Table 5: Requirement-to-Component Mapping

Identified Requirement	**Selected Component/Module**	**Function of the Component**
Measure soil moisture	Capacitive soil moisture sensor	Measures the moisture level in the soil and provides a signal to the microcontroller.
Process soil moisture readings	Arduino Uno	Receives and processes the readings from the soil moisture sensor.
Determine when irrigation is required	Arduino Uno + Embedded Firmware	Compares the measured moisture level with predefined thresholds.
Automatically start irrigation when soil is dry	Arduino Uno + Relay Module	The Arduino sends a control signal to the relay when irrigation is required.
Stop irrigation when sufficient moisture is reached	Arduino Uno + Relay Module	The Arduino switches the relay off, disconnecting power from the pump.
Display soil moisture information	LCD Display	Displays the moisture level and pump/system status to the farmer.
Continuously monitor soil condition	Soil Moisture Sensor + Arduino Uno	Periodically measures and processes soil moisture readings.
Supply water to crops automatically	Water Pump	Pumps water from the water source to the irrigation system.
Operate reliably	Arduino Uno + Appropriate Hardware	Provides stable control of the irrigation process when correctly installed and maintained.
Use electrical power efficiently	Embedded Firmware + Pump Control	Operates the pump only when irrigation is required and avoids unnecessary operation.

Identified Requirement	Selected Component/Module	Function of the Component
Respond quickly to dry soil	Soil Sensor + Arduino Firmware	Allows the system to detect changes in soil moisture and activate irrigation when required.
Be simple to operate	LCD Display + Automatic Control	Provides understandable information while reducing the need for manual irrigation control.
Operate safely	Relay Module + Appropriate Power Supply	Separates the microcontroller control circuit from the pump power circuit when correctly designed.
Provide consistent measurements	Capacitive Soil Moisture Sensor	Provides repeated soil moisture measurements used for irrigation decisions.

7. Overall System Operation

The complete operation of the Smart Irrigation System can be summarized in the following stages.

7.1 Stage 1: Soil Measurement

The soil moisture sensor is placed in the soil near the crops. It measures the amount of moisture present in the soil.

7.2 Stage 2: Data Processing

The sensor sends an electrical signal to the Arduino Uno. The microcontroller reads and processes the signal.

7.3 Stage 3: Irrigation Decision

The Arduino compares the sensor reading with the predefined moisture threshold.

For example:

$$\text{If Soil Moisture} < 30\%, \quad \text{Pump} = \text{ON} \tag{1}$$

and:

$$\text{If Soil Moisture} > 45\%, \quad \text{Pump} = \text{OFF} \tag{2}$$

These values are examples only. The actual thresholds should be determined through calibration and consideration of the crop and soil type.

7.4 Stage 4: Pump Control

When the soil is too dry, the Arduino sends a control signal to the relay module. The relay then switches the water pump on.

7.5 Stage 5: Irrigation

The pump supplies water to the crops through the irrigation pipes.

7.6 Stage 6: Automatic Shut-Off

As the soil becomes sufficiently moist, the sensor detects the change. The Arduino then switches the relay off and the pump stops.

7.7 Stage 7: Display

Throughout the process, the LCD displays the soil moisture condition and the status of the water pump.

8. Expected Benefits of the System

The proposed Smart Irrigation System is expected to provide the following benefits:

1. **Reduced water wastage:** Water is supplied only when the soil requires irrigation.
2. **Reduced labour costs:** The farmer does not need to continuously monitor and manually operate the irrigation system.
3. **Improved irrigation timing:** The system responds to the actual moisture condition of the soil.
4. **Improved crop production:** Maintaining appropriate soil moisture can support healthier crop growth.
5. **Efficient electrical power usage:** The water pump operates only when irrigation is required.

6. **Continuous monitoring:** The farmer can view the soil condition using the LCD display.
7. **Automation:** The system reduces the need for continuous human intervention.

9. Safety and Reliability Considerations

Because the proposed system combines electronics, electricity and water, appropriate safety measures are required.

- The pump should be powered using a suitable power supply.
- The pump must not be connected directly to an Arduino output pin.
- The relay should be appropriately rated for the pump voltage and current.
- Electrical components should be protected from water and excessive moisture.
- Appropriate enclosures should be used for outdoor installations.
- Wiring should be properly insulated and secured.
- The system should be tested before deployment on a large farm.
- Sensor readings should be calibrated for the actual soil and crop conditions.

For a real farm installation, electrical protection and installation should be performed according to appropriate safety standards and by a suitably qualified person where necessary.

10. Possible Future Improvements

The initial system can be expanded in the future.

Possible improvements include:

- Replacing the Arduino Uno with an ESP32.
- Adding Wi-Fi connectivity.
- Providing smartphone monitoring.
- Sending notifications when irrigation occurs.
- Recording soil moisture data over time.
- Adding multiple moisture sensors for different areas of the farm.

- Adding automatic weather information.
- Integrating a solar power system.
- Adding a water tank level sensor.
- Developing a web or mobile dashboard.

These improvements could make the system more suitable for large-scale agricultural applications.

11. Conclusion

The proposed Smart Irrigation System provides an automated solution to the farmer's irrigation challenges. The system uses a soil moisture sensor to measure the condition of the soil and sends the information to an Arduino Uno microcontroller.

The microcontroller processes the sensor reading and determines whether irrigation is required. When the soil is too dry, the Arduino activates a relay module, which switches on the water pump. The pump supplies water to the crops until the required moisture condition is reached. The pump is then switched off automatically.

An LCD display provides the farmer with information about the soil moisture and pump status. The proposed system can reduce water wastage, reduce unnecessary labour, improve irrigation timing and support better crop production.

The initial design provides a practical embedded-system solution that can later be expanded with technologies such as Wi-Fi, mobile monitoring, cloud data storage and solar power.

12. Presentation Summary

For the group presentation, the project can be divided into the following sections:

Table 6: Suggested Group Presentation Structure

Member	Section	Main Points
1	Introduction and Requirements	Explain the farmer's problem, system objectives and functional/non-functional requirements.
2	Components	Explain the Arduino, soil moisture sensor, relay, pump, power supply and LCD.
3	Block Diagram	Explain the data flow and control signals between the system components.
4	Requirement Mapping	Explain how the requirements are implemented by the selected components.
5	System Operation and Conclusion	Explain the complete operation, benefits, safety and future improvements.